Zeta-Function Regularization

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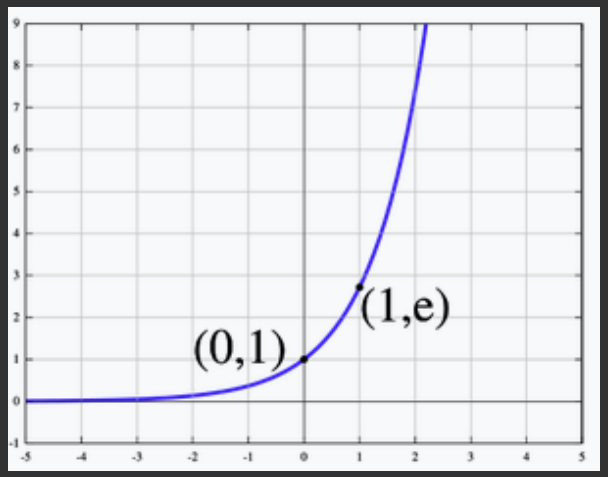
Table of Contents

1. Analytic Continuation
   - Definition
   - Uniqueness
2. The Riemann Zeta
   - Analytic Continuation to $\mathbb{C}$
3. Zeta Function Regularization
   - Path Integrals in Curved Spacetime
   - Casimir Effect
4. References

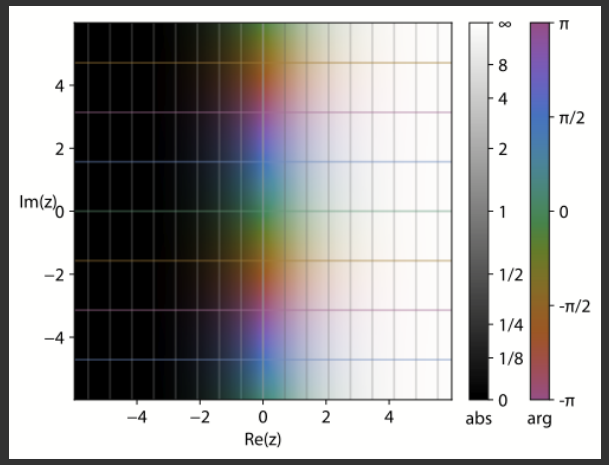
Analytic Continuation

Introduction

- Given an analytic function on some domain D_1, we want to expand it to a larger region D_2 such that $D_1 \subseteq D_2$.
- This is called analytic continuation.
- For example, the complex exponential $e^z = e^x (\cos x + i \sin y)$ is the analytic continuation of the real-valued exponential e^x.



(a) Real exponential



(b) Complex exponential

Analytic Continuation

- Specifically, let D_1, D_2 be two domains in $\mathbb{C}$. Suppose that f_1 is analytic in D_1, and f_2 is analytic in D_2.
- If $f_2(z) = f_1(z)$ for all $z \in D_1 \cap D_2$, then f_2 is called the analytic continuation of f_1 into D_2.

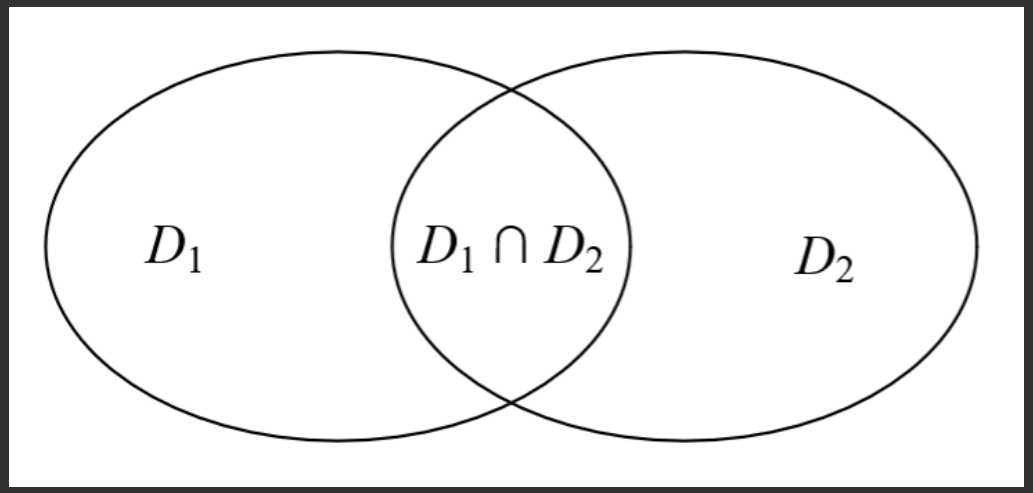


Figure 1: domains D_1 and D_2

Uniqueness (part 1)

Theorem: Let $f : D(\omega, R) \to \mathbb{C}$ be an analytic function from the disk centered at ω with radius R to $\mathbb{C}$. Assume that f is not identically 0. Then there exists $0 < r \leq R$ such that $f(z) \neq 0$ whenever $0 < |z - w| < r$.

Proof: If $f(z) \neq 0$, we can form a neighborhood by continuity of f.

If $f(z) = 0$, we can represent it with a power series $f(z) = \sum_{n=0}^{\infty} a_n (z - w)^n$ with $a_0 = 0$. Since f is not identically 0, some coefficients will be nonzero. Then for some r, $f(z)$ is nonvanishing on the disk $0 < |z - w| < r$.

I.e. zeros are isolated. We can form a neighborhood around each zero that contains no other zeros.

Uniqueness (part 2)

Theorem: Let f, g be analytic functions on the domain $D \subset \mathbb{C}$. If the set $E = \{z \in D \mid f(z) = g(z)\}$ contains an accumulation point, then $f(z) = g(z)$ for all $z \in D$.

Proof: The function $h(z) = f(z) - g(z)$ is analytic. Let $\omega \in E$ be the accumulation point of E. Then h vanishes on some disk D_ω.

Let $a \in D \setminus D_\omega$. Let $\gamma : [0,1] \to D$ be a path with $\gamma(0) = \omega$ and $\gamma(1) = a$. Define t_0 as the "boundary" of D_ω,

$$t_0 = \sup\{t \in [0,1] \ \mid \ h(\gamma(s)) = 0, s \in [0,t]\}$$

h vanishes for some points in a neighborhood of $\gamma(t_0)$, which violates the previous theorem. So h must be identically 0, and $f(z) = g(z)$ for all $z \in D$.

Example: Grandi's Series

- Analytic continuation gives weird results, if taken at face value.
- Consider the following series:

$$f(x) = \sum_{n=0}^{\infty} (-x)^n = \frac{1}{1+x} \ , \quad |x| < 1$$

- We analytically continue the function to $\mathbb{C}$, saying $f(x) = (x+1)^{-1}$ for all $x \in \mathbb{C}$.
- Setting $x = 1$, we get $1 - 1 + 1 - 1 + \cdots = \frac{1}{2}$. Obviously false.[1]
- This is called Grandi's series.

[1] Or is it?

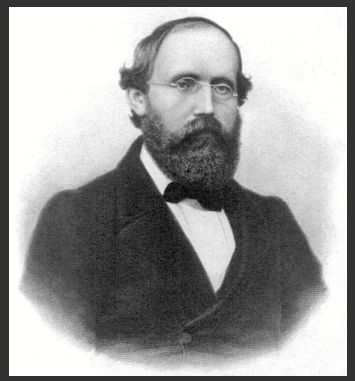
The Riemann Zeta

The Zeta Function

- Equally famous among mathematicians and middle schoolers.
- There's a lot of variants, but the Riemann zeta function is

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}\ , \quad \mathrm{Re}(s) > 1$$

- This only converges if the real part of s is > 1 (p-test).
- Riemann's work was to generalize the zeta function to complex numbers $s = \sigma + it$, then analytically continue it to $\mathbb{C}$.
- We'll reconstruct those steps.



The Functional Equation (part 1)

$$\zeta(s) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{x^{s-1}}{e^x - 1}\, dx \ , \quad \mathrm{Re}(s) > 1$$

Start from the integral and express it as a geometric series:

$$\int_0^\infty \frac{x^{s-1}}{e^x - 1}\, dx = \int_0^\infty \frac{x^{s-1} e^{-x}}{1 - e^{-x}}\, dx = \int_0^\infty \sum_{n=0}^\infty x^{s-1} e^{-(n+1)x}\, dx$$

Switch the sum and integral, then substitute $u = (n+1)x$.

$$\int_0^\infty \frac{x^{s-1}}{e^x - 1}\, dx = \sum_{n=0}^\infty \int_0^\infty \frac{u^{s-1}}{(n+1)^s} e^{-u}\, du$$

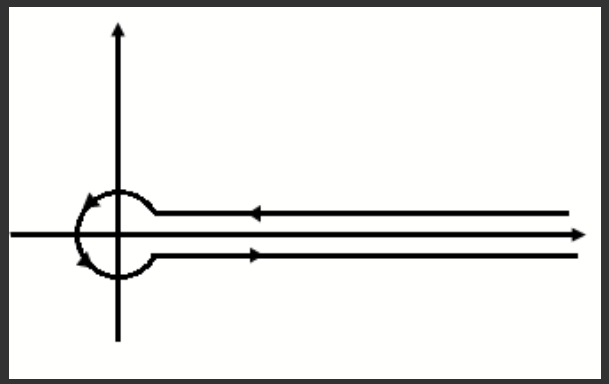
Substitute the Gamma function to get

$$\int_0^\infty \frac{x^{s-1}}{e^x - 1}\, dx = \sum_{n=0}^\infty \frac{\Gamma(s)}{(n+1)^s} = \Gamma(s)\zeta(s)$$

The Functional Equation (part 2)

Consider the Hankel integral:

$$\int_C \frac{(-z)^{s-1}}{e^z - 1}\, dz$$



From the bottom, $(-z) \approx xe^{-i\pi}$, and from the top, $(-z) \approx xe^{i\pi}$.

$$\int_C \frac{(-z)^{s-1}}{e^z - 1}\, dz = \left(e^{-i\pi(s-1)} - e^{i\pi(s-1)}\right) \int_0^\infty \frac{x^{s-1}}{e^x - 1}\, dx$$
$$= 2i \sin(\pi s)\, \Gamma(s)\zeta(s)$$

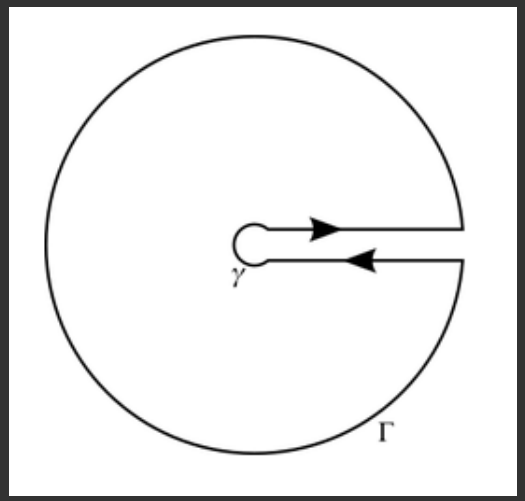
Use the reflection property $\Gamma(s)\Gamma(1-s) = \pi/\sin(\pi s)$ to get

$$\boxed{\zeta(s) = \frac{\Gamma(1-s)}{2\pi i} \int_C \frac{(-z)^{s-1}}{e^z - 1}\, dz}$$

The Functional Equation (part 3)

Consider the keyhole integral

$$\int_K \frac{(-z)^{s-1}}{e^z - 1}\, dz$$



This has singularities at $s = 2\pi i n$ with residues $(2\pi i n)^{s-1}$.

By Cauchy's residue theorem,

$$\begin{aligned}
\int_K \frac{(-z)^{s-1}}{e^z - 1}\, dz &= 2\pi i \left(\sum_{n=1}^{\infty} (2\pi i n)^{s-1} + \sum_{n=1}^{\infty} (-2\pi i n)^{s-1} \right) \\
&= 2\pi i (2\pi)^{s-1} \left(e^{i\frac{\pi(s-1)}{2}} - e^{-i\frac{\pi(s-1)}{2}} \right) \sum_{m=1}^{\infty} \frac{1}{m^{1-s}} \\
&= 2\pi i (2\pi)^{s-1} \cdot 2 \sin\left(\frac{\pi(s-1)}{2} \right) \zeta(1-s)
\end{aligned}$$

The Functional Equation (part 3)

From previous results:

$$\zeta(s) = \frac{\Gamma(1-s)}{2\pi i} \int_C \frac{(-z)^{s-1}}{e^z - 1}\, dz$$

$$\int_K \frac{(-z)^{s-1}}{e^z - 1}\, dz = 2\pi i (2\pi)^{s-1} \cdot 2 \sin\left(\frac{\pi(s-1)}{2}\right) \zeta(1-s)$$

Equating the integrals, we get the functional equation of the Riemann Zeta. Note that we've found analytically continued values for $\zeta(1-s)$ in terms of $\zeta(s)$.

$$\boxed{\zeta(s) = \Gamma(1-s)\pi^{s-1}2^s \cos\left(\frac{\pi(s-1)}{2}\right) \zeta(1-s)}$$

Values of the Analytically Continued Zeta

$$\zeta(s) = \Gamma(1-s)\pi^{s-1}2^s \cos\left(\frac{\pi(s-1)}{2}\right)\zeta(1-s)$$

$$\zeta(1-x) = 2^{1-x}\pi^{-x}\cos\left(\frac{\pi x}{2}\right)\Gamma(x)\zeta(x)\ , \quad x = 1-s$$

- Using the functional equation, we can find analytically continued values of the zeta function.
- For example,[2]

$$\zeta(-1) = -\frac{1}{2\pi^2}\cdot\zeta(2) = -\frac{1}{12}$$

[2] This is where the "proof" that $1+2+3+\cdots = -\frac{1}{12}$ comes from.

Zeta Function Regularization

Path Integrals in Curved Spacetime

- One of the earliest uses of zeta function regularization was a paper by Hawking in 1977.
- The idea is to use a spectral zeta function. For an operator A with eigenvalues λ,

$$\zeta_A(s) = \sum_{n=1}^{\infty} \frac{1}{\lambda_n^s}$$

- Note that

$$\left.\frac{d}{ds}\zeta_A(s)\right|_{s=0} = -\sum_{n=1}^{\infty} \log \lambda_n = -\log \det A$$

$$\boxed{\det A = e^{-\zeta_A'(0)}}$$

Path Integrals in Curved Spacetime

- If we come across an operator with an infinite amount of eigenvalues,[3] we regularize it with the previous result.
- Simply substitute $\det A = e^{-\zeta_A'(0)}$, where ζ_A is the spectral zeta.
- $\zeta_A(s)$ can be found in terms of the Riemann zeta. For example, Hawking found (for his A)

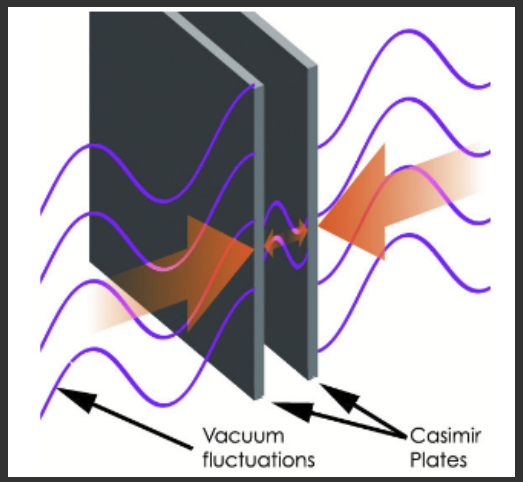
$$\zeta_A(s) = -\frac{8\pi V}{(2\pi)^3}(2\pi\beta^{-1})^{3-2s} \cdot \zeta(2s-3)$$

$$\det A = e^{-\zeta_A'(0)}$$

[3] Specifically: infinite eigenvalues can occur on infinite-dimensional Hilbert spaces.

The Casimir Effect

- Another application is the Casimir effect, where a force arises between two uncharged plates in a vacuum.
- The expected vacuum energy $\langle E \rangle$ is unbounded.



$$\frac{\langle E \rangle}{A} = \int \frac{1}{(2\pi)^2} \, dk_x dk_y \sum_{n=1}^{\infty} \omega_n$$

- But we tweak the expression to force in a zeta:

$$\frac{\langle E \rangle}{A} = \lim_{s \to 0} \int \frac{1}{(2\pi)^2} \sum_{n=1}^{\infty} \omega_n |\omega_n|^{-s} \, dk_x dk_y$$

The Casimir Effect

- After cumbersome substitutions, we get

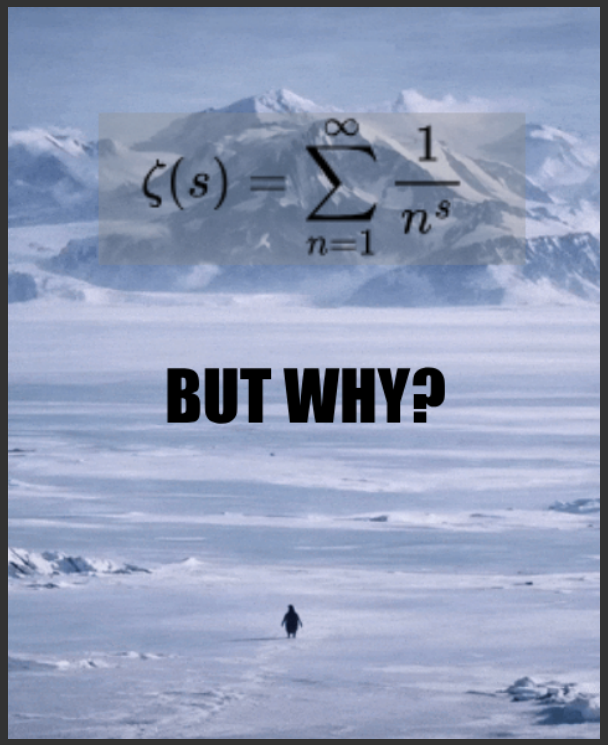
$$\frac{\langle E \rangle}{A} = \lim_{s \to 0} \int \frac{1}{(2\pi)^2} \sum_{n=1}^{\infty} \omega_n |\omega_n|^{-s} \, dk_x dk_y$$
$$= \lim_{s \to 0} -\frac{1}{2r^{3-s}} \frac{\pi^{2-s}}{3-s} \sum_{n=1}^{\infty} |n|^{3-s} = -\frac{1}{2r^3} \frac{\pi^2}{3} \zeta(-3)$$

- Using the analytically continued value of $\zeta(-3)$,

$$\boxed{\frac{\langle E \rangle}{A} = -\frac{\pi^2}{6r^3} \cdot \frac{1}{120} = -\frac{\pi^2}{720r^3}}$$

- This is an odd result: a nonzero force between non-interacting plates.

Caveats

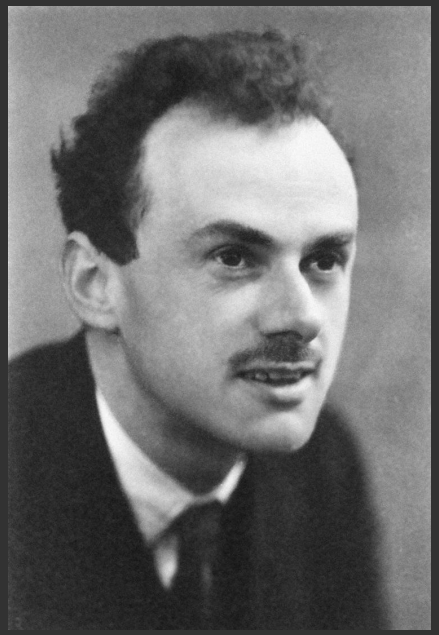


- Blindly applying zeta function regularization gives unexpected results, e.g. $1 + 2 + 3 + \cdots = -\frac{1}{12}$
- Why do we use an obviously incorrect result?
- Short answer: it works. We don't know why.

Technically, it's mathematically unsound to "do away" with infinities. Physicists do it anyway because it fits with experiments.

Alternatives

- Other regularization schemes exist.
- Well-known ones include Pauli-Villars regularization and dimensional regularization.
- But the fact that regularization needs to exist is controversial.



Dirac opposed regularization.

The need for any regularization scheme at all motivates a search for new physics beyond the Standard Model.

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